

Observability in DevOps:

Tools and Techniques

Agenda

- 01 Introduction
- 02 Monitoring and Observability
- 03 Observability Tools in DevOps
- 04 Best practices
- 05 Case study and demo

01 Introduction

Who am I?

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What (am I)?

DevOps (?), Cloud Engineer, Automation Engineer, Platform Engineer , Infrastructure Engineer, SW Engineer, QAE, Prompt Manager,
Jack of all trades

What (do I do)?

It depends.

What am I going to talk about?

DevOps

Monitoring

Observability

Tools

Practices

01 Introduction

DevOps?

CULTURE, WAY OF WORKING

DevOps Role?

Should not exist?

But it does. Preacher of DevOps, Enabler, CI/CD, Tools master, Cloud SME, Automation guru

What does a DevOps do?

Automates

Application Lifecycle

Environments

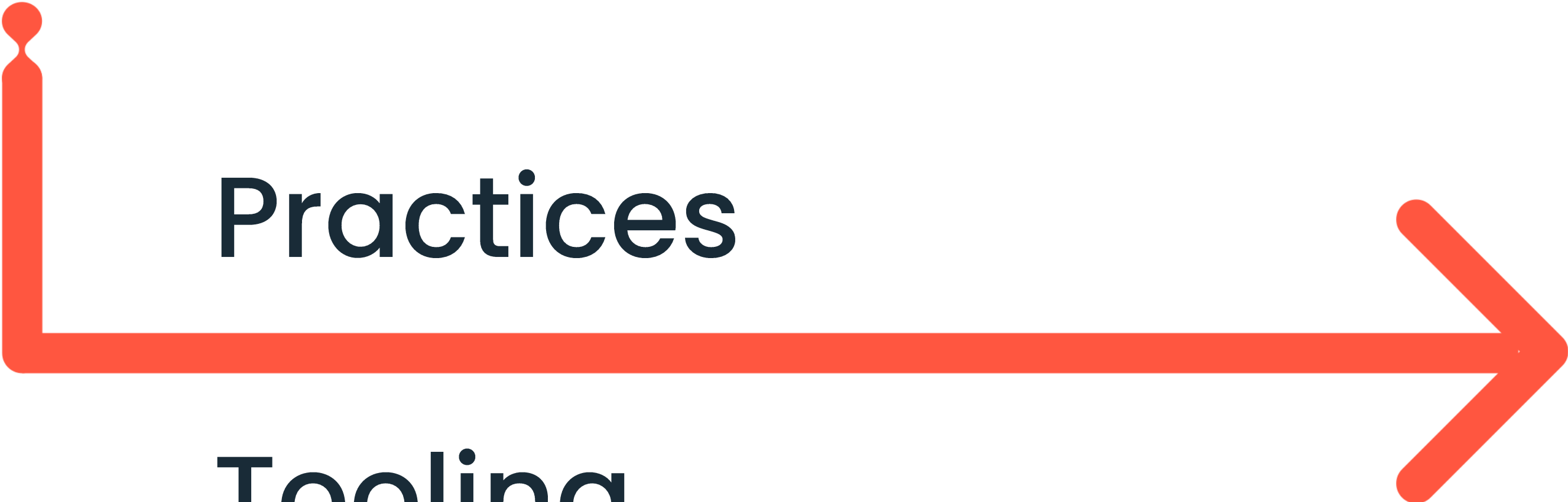
Operate and strengthens the solution

Monitoring, Alerting, Reporting

Optimizations, Security, Patching

02 ∞ Monitoring and Observability

Monitoring



Practices

Tooling

Observability.

Monitoring

Actively tracking and measuring by collecting data and then visualizing in a way that provides insight.

Reactive and responds to “**What happened?**”

Key words

1. **Metrics** – numerical measure (CPU, mem usage)
2. **Logs** – historical records
3. **Traces** – the flow of the individual requests
4. **Alerting** – alert on important events
5. **Dashboards** – visual representation

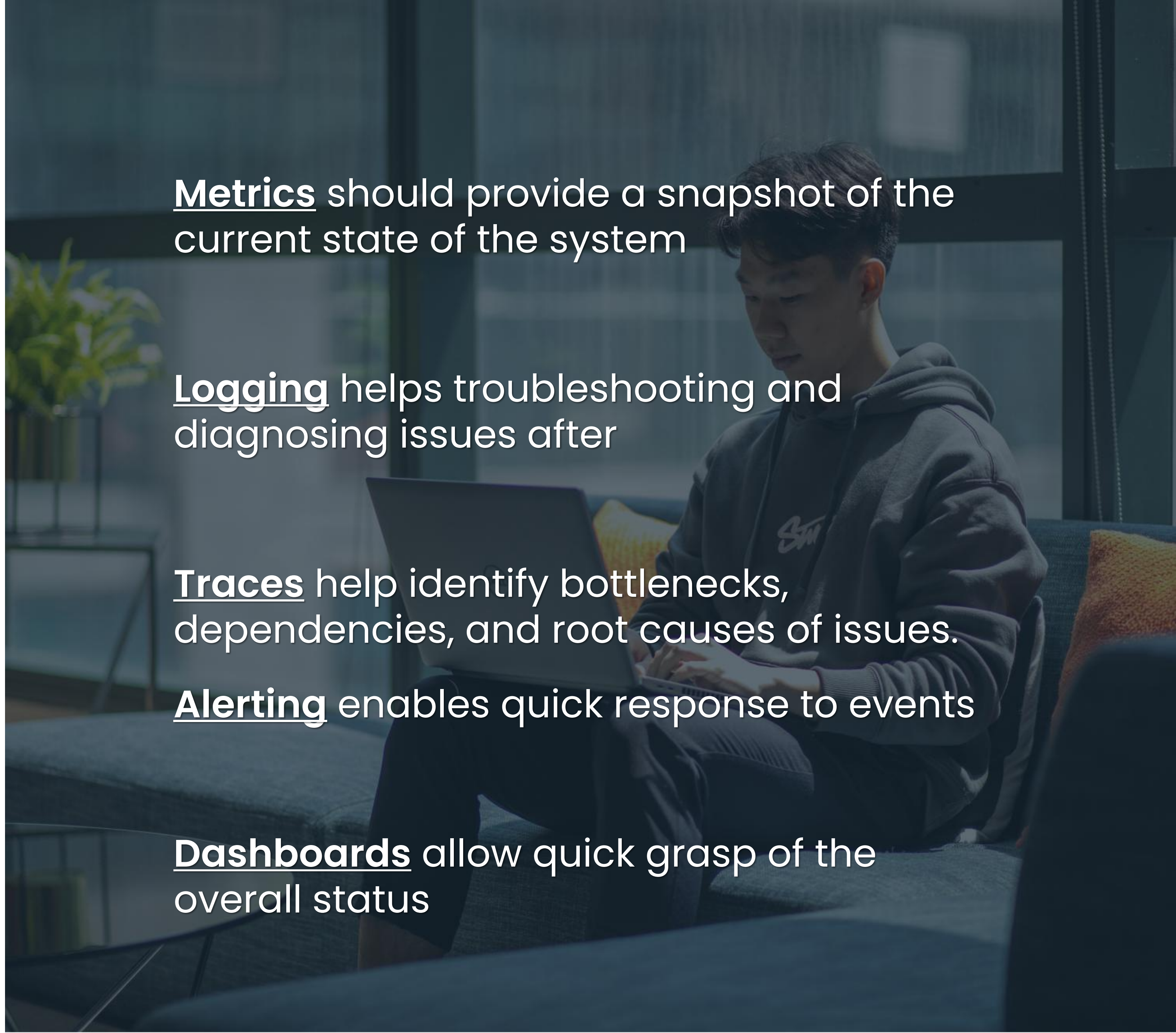
Metrics should provide a snapshot of the current state of the system

Logging helps troubleshooting and diagnosing issues after

Traces help identify bottlenecks, dependencies, and root causes of issues.

Alerting enables quick response to events

Dashboards allow quick grasp of the overall status



Observability

Monitoring+++

Is about gaining deep understanding of how the system works by exploring the data made available by monitoring

Proactive and responds to the “**Why it happened?**”

Key aspects

- **High Cardinality Data:** detailed info about individual transaction or event
- **Distributed tracing:** ability to trace a request end to end
- **Correlation:** correlate data from different sources to provide a holistic view of the system behavior
- **Querying and exploration :** Observability tools enable users to explore and analyze data real-time

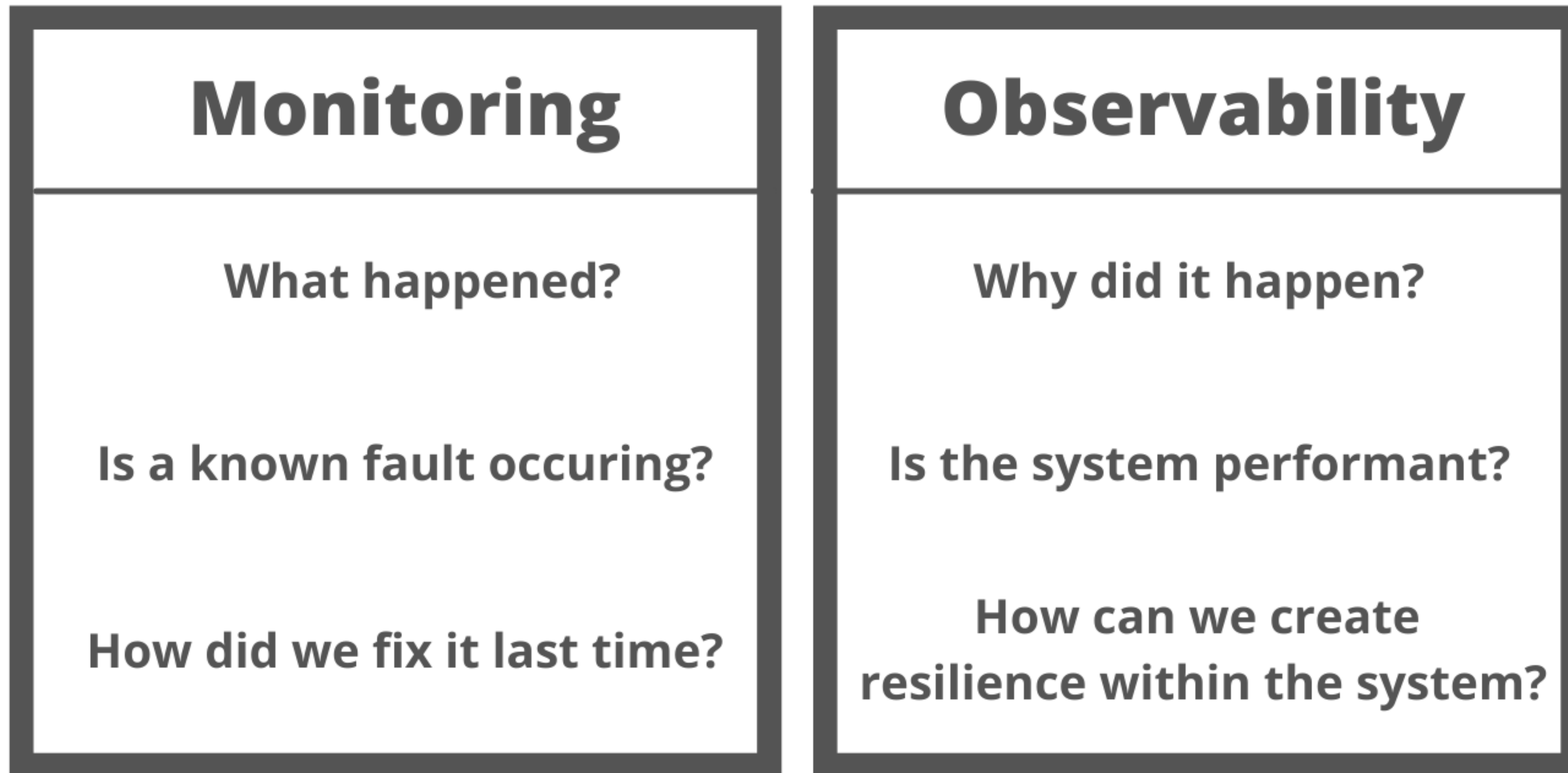
High Cardinality Data allows exploring the detail of a request

Distributed Tracing make it easy to identify bottlenecks and troubleshoot issues in Distributed Systems

Correlation helps make sense of the data available

Q&E enables navigation of the “lakes” of data

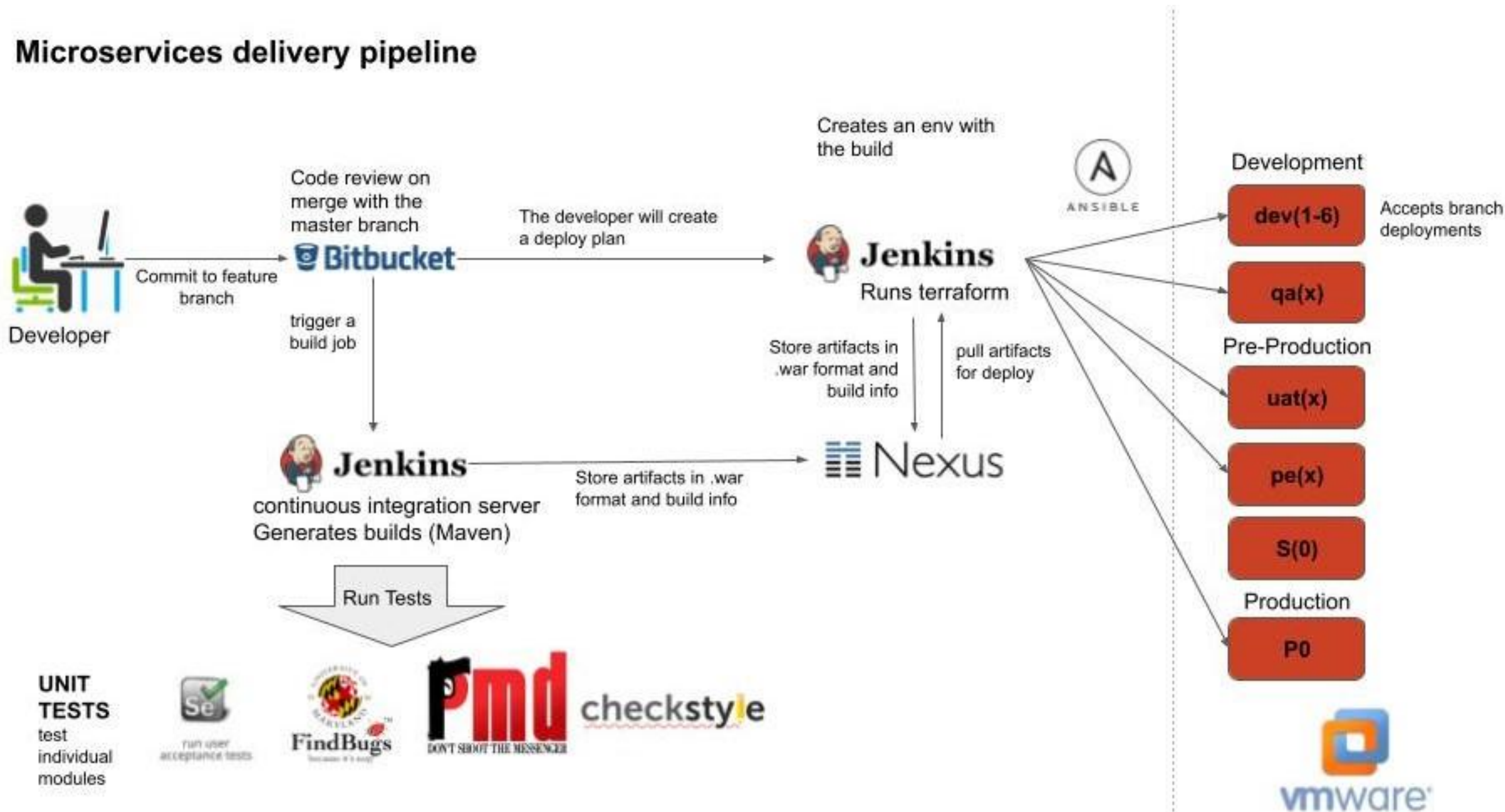
Monitoring vs Observability



The source link for the image above is [here](#)

Metrics from a DevOps perspective

Microservices delivery pipeline



Metrics from a devops perspective

What extra data we have from a DevOps perspective?

- Change management
 - a) IaasC runs
 - b) Security compliance – dynamic and static scanners
 - c) Release management – Speed? Impact on the system?
 - d) Quality of the code – code quality checkers, unit tests, integration tests, performance tests
- Application management
 - a) Backups
 - b) Log retention
 - c) Security patching
 - d) Automatic response procedures (DR, failover, response time)

03 DevOps Tooling

- Diverse
 - Over 30 tools, providers and services some of them with a rich ecosystem (AWS, Azure, etc.)
- Let's explore [DevOps Roadmap](#)
- Monitoring and Observability are key to DevOps both as a role and as Culture

Observability Tools

Popular all-in-one tools

- Expensive
- Most of them have a free version
- The free version is usually specialized
- Cloud provider specific



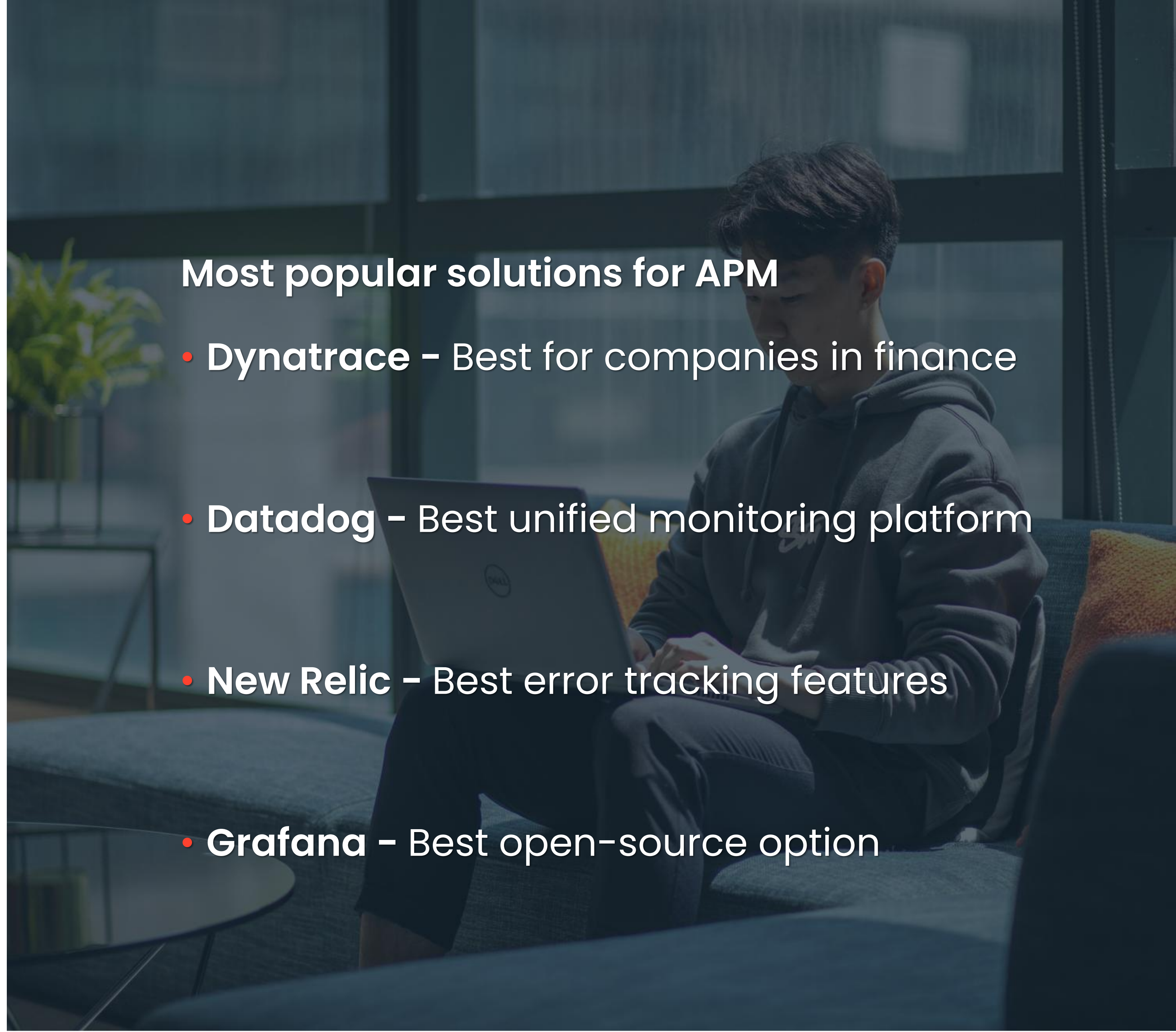
Observability Tools

Popularity contest categories:

- Type of business
- Distribution
- Costs
- Integrations
- Ease of migration
- Scalability
- Features

Most popular solutions for APM

- **Dynatrace** – Best for companies in finance
- **Datadog** – Best unified monitoring platform
- **New Relic** – Best error tracking features
- **Grafana** – Best open-source option



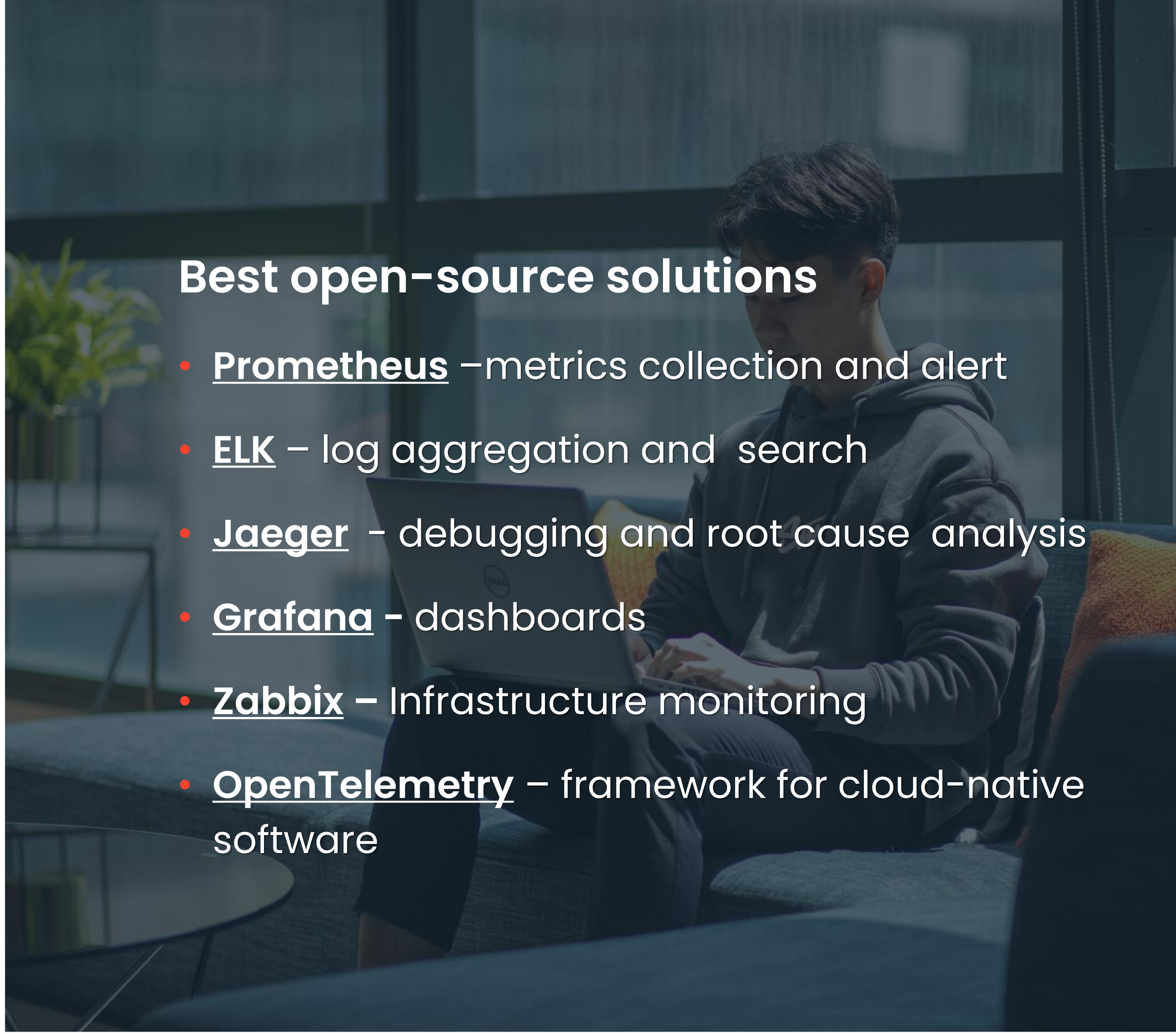
Observability Tools

Opensource popularity contest:

- Metrics
- Log aggregation
- Alerts
- Debugging
- Root cause analysis
- Dashboards
- Integrations

Best open-source solutions

- Prometheus – metrics collection and alert
- ELK – log aggregation and search
- Jaeger – debugging and root cause analysis
- Grafana – dashboards
- Zabbix – Infrastructure monitoring
- OpenTelemetry – framework for cloud-native software



04 Best practices

Learn it!

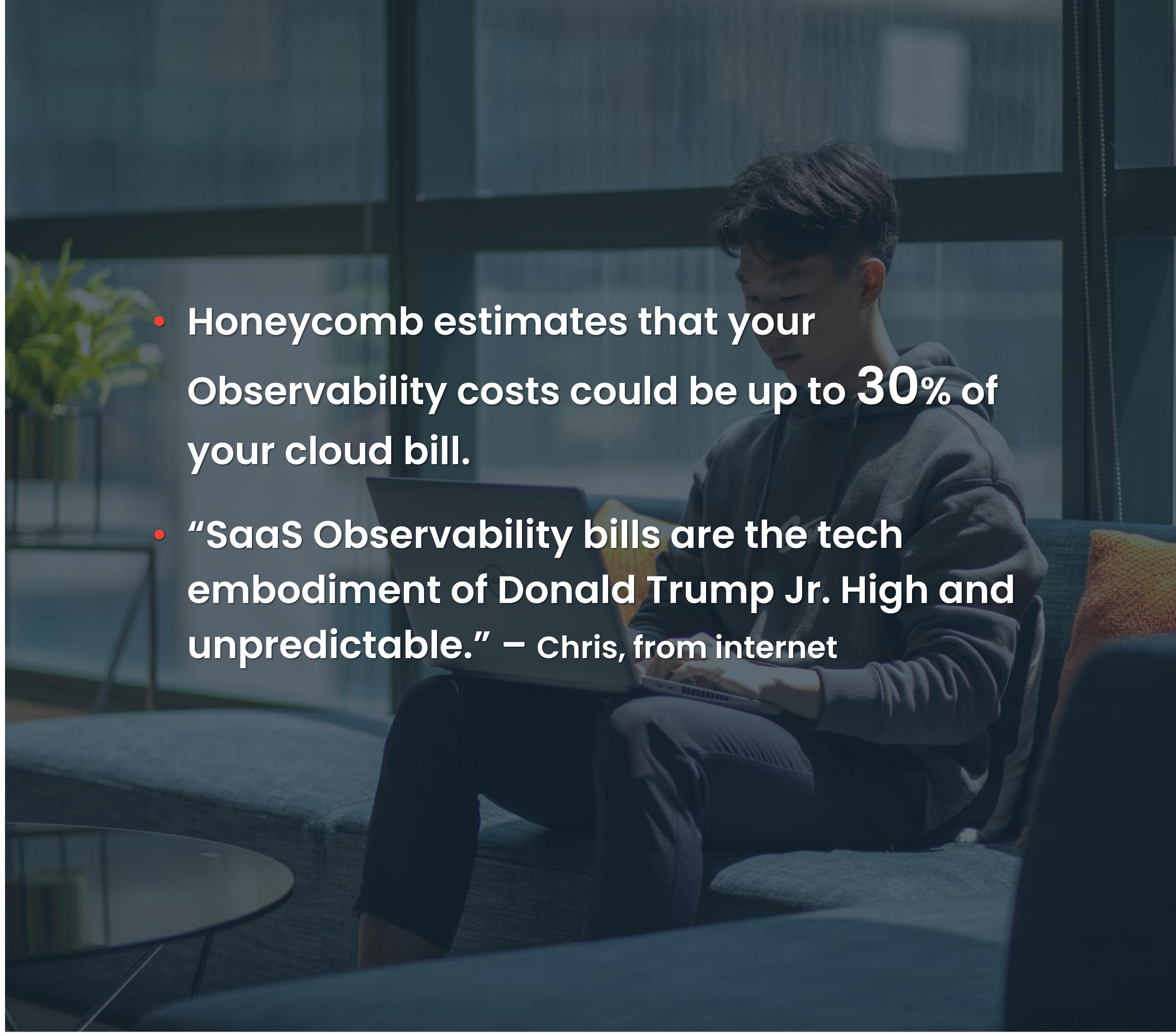
Build it!

Use it!

Best practices, part 1

1. Beware of costs!
2. Establish the scope
3. Observe only relevant information
4. Context is King! Collect lots of it
5. Put alerts only for critical events
6. Integrate with collaboration tools
7. Design the architecture and application with observability in mind
8. Beware of added complexity

- Honeycomb estimates that your **Observability costs could be up to 30% of your cloud bill.**
- **“SaaS Observability bills are the tech embodiment of Donald Trump Jr. High and unpredictable.” – Chris, from internet**



Best practices, part 2

- 9. Observability to optimize costs
- 10. Don't rely on default dashboards! Make them relevant
- 11. Integrate with automated remediation systems wherever possible
- 12. Use standardized, relevant names!
- 13. Generate data graphs that are easy to understand
- 14. Adapt tools and configs to your use case – some tools are more resource intensive than others
- 15. Data retention strategies



Best practices, part 3

1. Integrate “DevOps data” in the overall Observability system.

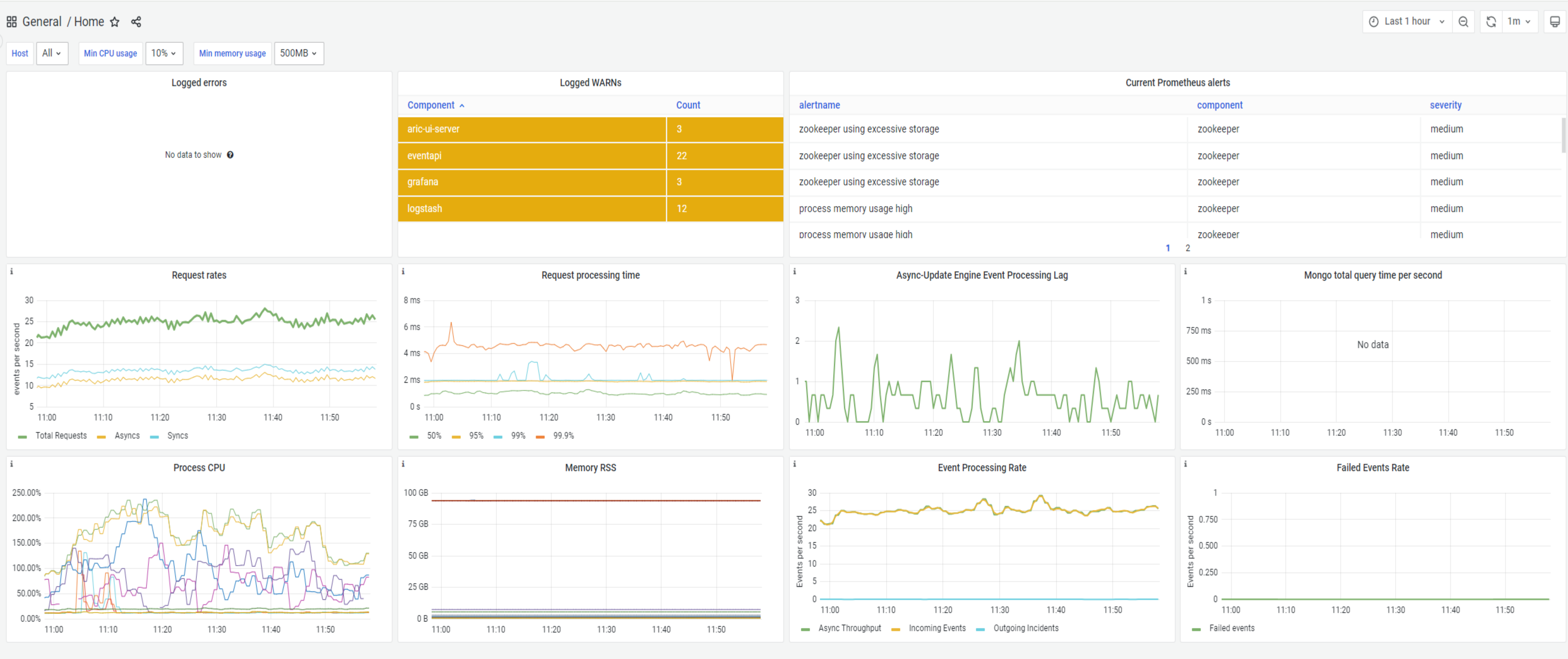
Why? **End to end observability, ownership and responsibility**

2. How to do that?
 - a) Make the data easily available
 - b) Integrate notification systems with the pipelines
 - c) Alert on automatic events (DRs, failovers, rollouts, rollbacks, etc.)
3. Status as a code – desired, existent and delta
4. Use the same practices as for the APM Observability

05 Demos

1. Grafana
2. Short ELK video demo

Demo Grafana



Datadog

Application Uptime

[DE] [SHOIST] Shipping-produce

Past 7 Days

100.00%

100% (3h 21m) budget

98% target

Total Misconfigs

14.7k

Critical Misconfigs

3

System Load by Availability Zone

Updated < 1 min ago

as of Wed 17 Jan 11:44

Lambda Invocations Monitors

Alert 14

Warn 1

OK 12

Lambda Errors

59

0 errs

Threats by severity

RDS DBs with Highest Connections

hetansh-test... N/A oreo +7

Running EKS pods per node

Lambda Invocations

Service Dependencies (Payments)

Checkout duration per cluster

Top 10 high latency customers

8.30	Martinez PLC
7.77	Campbell-Curry
2.78	Davis-Davis
2.49	Henry-Harris
2.20	LA Gear

Log Stream

AWS Dashboard

Share Configure Edit Dashboard

Saved Views

name Store_id service coffee-house

1d Past 1 Day

Navigation icons



Application Uptime

[DE] [SHOIST] Shipping-produce

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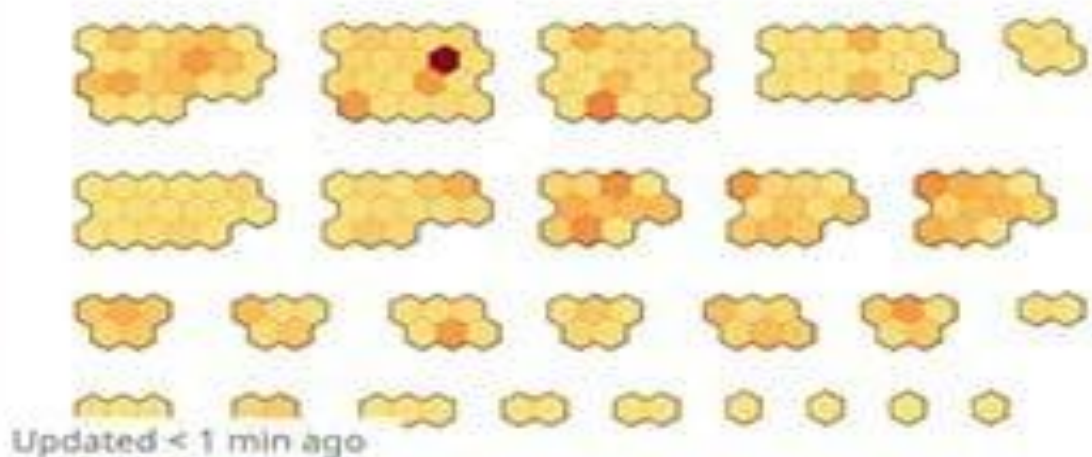
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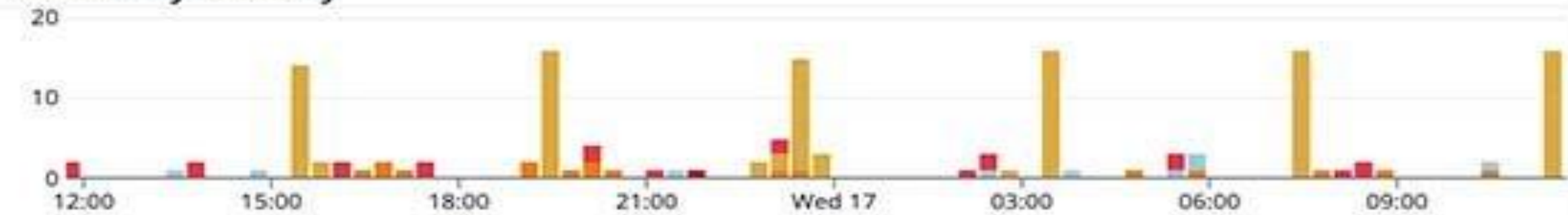
Lambda Invocations Monitors



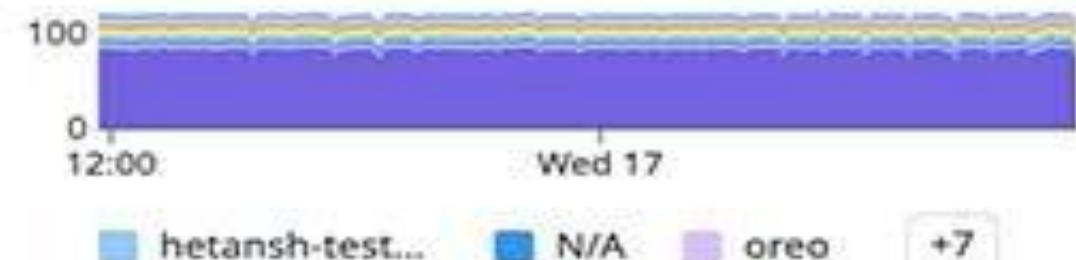
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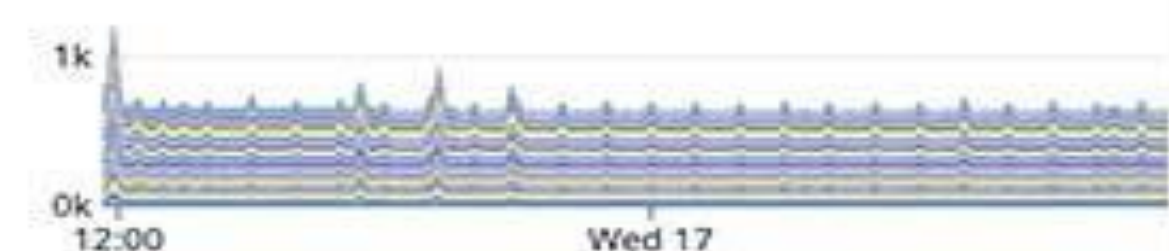
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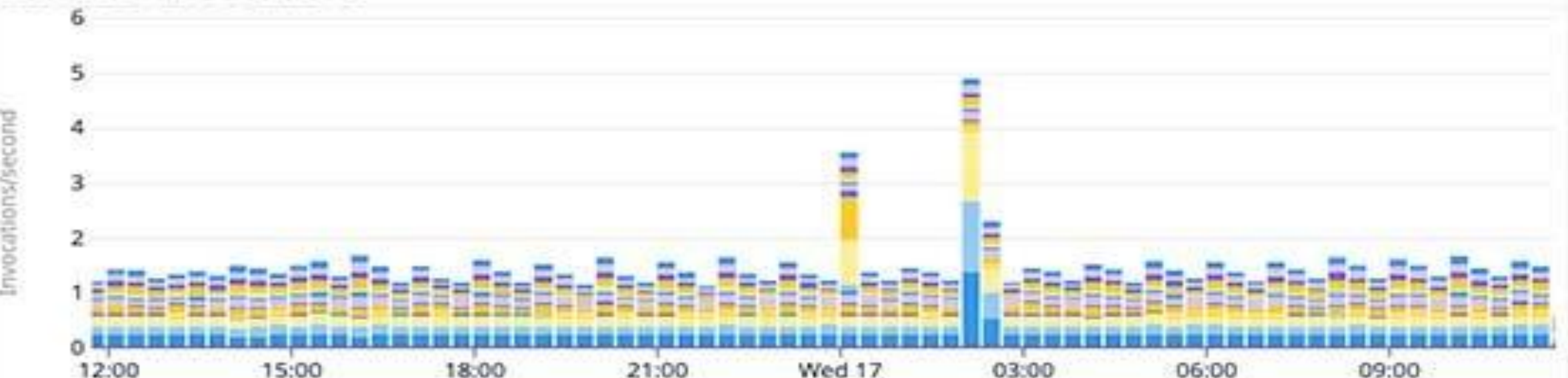
RDS DBs with Highest Connections



Running EKS pods per node



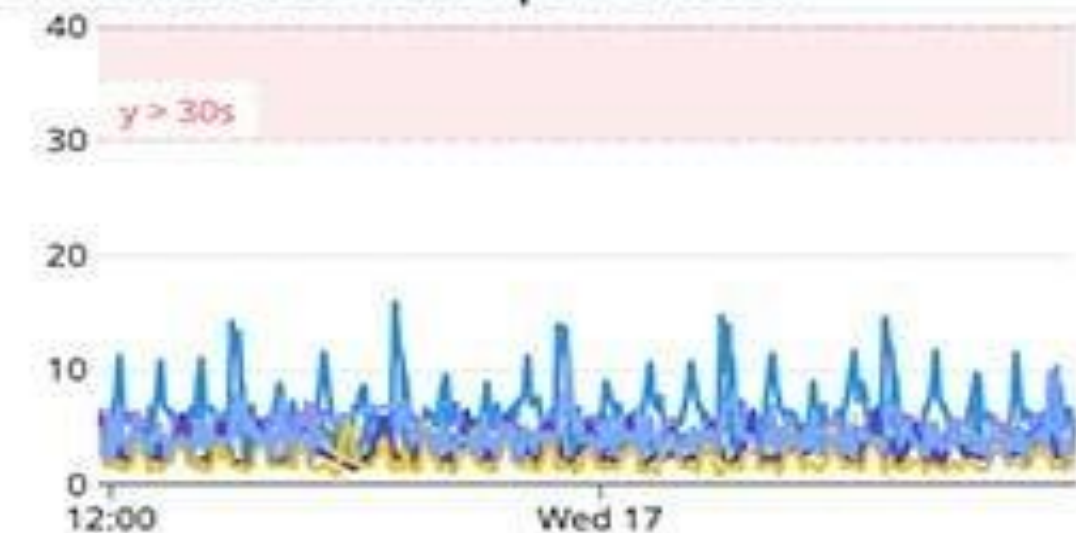
Lambda Invocations



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Q&A

Thank *you*!